

Project 4

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Design

a. Circuit Schematic.

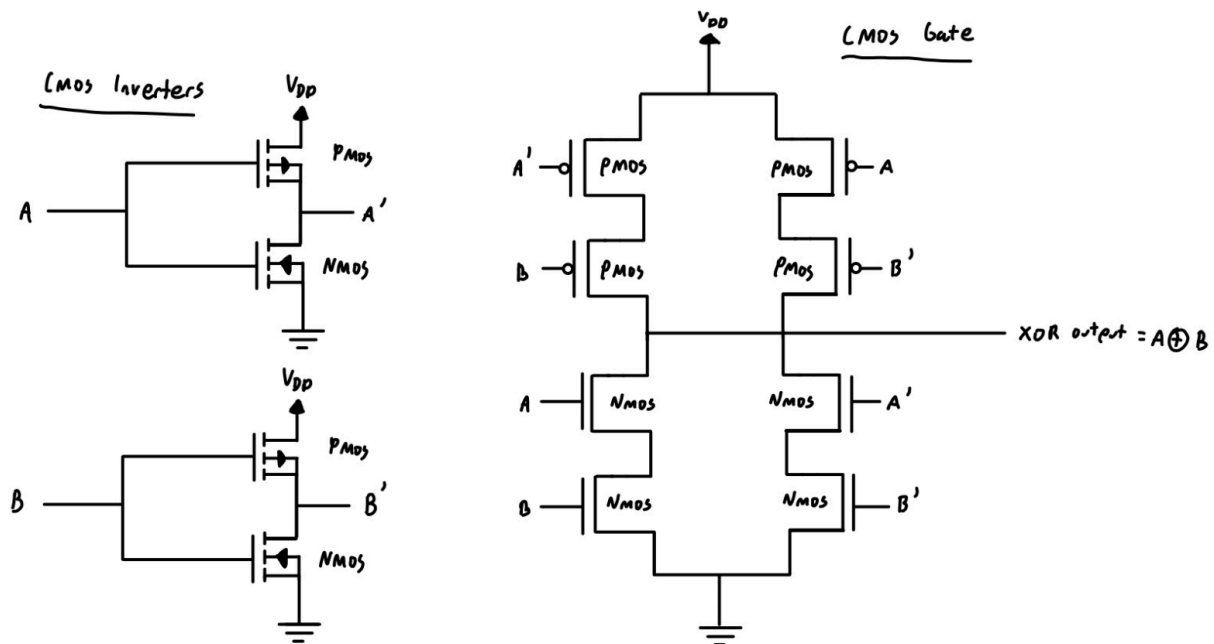
To build the XOR gate, 6 NMOS and 6 PMOS were used. This is necessary to create both the inverter and XOR gate in CMOS. We can use DeMorgan's theorem to explain the design better.

Boolean expression for a XOR gate:

$$\begin{aligned}
 Y &= A \oplus B \\
 Y &= A'B + B'A \\
 Y &= ((A'B + B'A)')' \\
 Y &= (AB + A'B')'
 \end{aligned}$$

Keeping this in mind, the inverter and CMOS gate can be created as a system of pull-down and pull-up networks.

Figure 1: CMOS gate and inverters for A' and B'



b. Ideal Sizing

To minimize delay in a CMOS XOR gate, the pull-up and pull-down networks must be symmetrically balanced. The overall propagation delay of the circuit is governed by two primary factors: the load capacitance and the average effective resistance during state transitions. To achieve this balance, we use a standard unit CMOS inverter as our baseline. Because NMOS electron mobility is 2.5 times higher than PMOS hole mobility, the reference PMOS transistors must be sized 2.5 times larger than the NMOS transistors to match resistance.

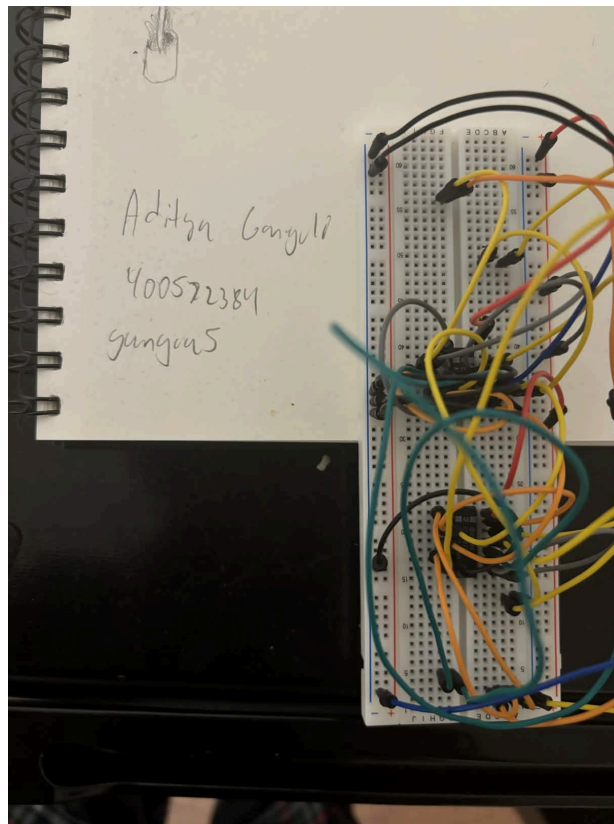
When applying this to the XOR gate, the optimal dimensions can be systematically calculated to minimize delay by identifying the longest series path, which represents the worst-case scenario. In this XOR design, the worst-case path involves two transistors in series, which doubles their effective resistance. Because resistance is inversely proportional to physical size, each transistor in these critical paths must be sized twice as large as those in the reference inverter. Ultimately, compensating for this series resistance ensures the gate matches the reference delay and preserves the ideal 2.5:1 PMOS-to-NMOS size ratio throughout the design.

- c. Explain whether you can implement the ideal sizing in your hardware design. If not, explain why not, and qualitatively explain the impact on circuit performance.**

It is not possible to implement the calculated ideal sizing in this specific hardware design. The primary limitation is that the circuit is constructed using two pre-packaged IC MOSFETs (the CD4007CN) that contain 6 PMOS and 6 NMOS each. The CD4007CN ICs feature fixed transistor dimensions (W/L). Without the ideal 2.5:1 ratio, the pull-up and pull-down network resistances are unbalanced sizing-wise. The relatively undersized PMOS transistors create a higher effective resistance in the PUN. This qualitatively causes asymmetric performance, as the higher resistance bottlenecks the charging process, resulting in a slower rise time (t_{rise}) and a longer low-to-high propagation delay (τ_{PLH}) compared to the pull-down transitions. Ultimately, this asymmetry prevents the gate from achieving its theoretical minimum delay.

Testing

Figure 2: Physical Circuit of the CMOS XOR Gate



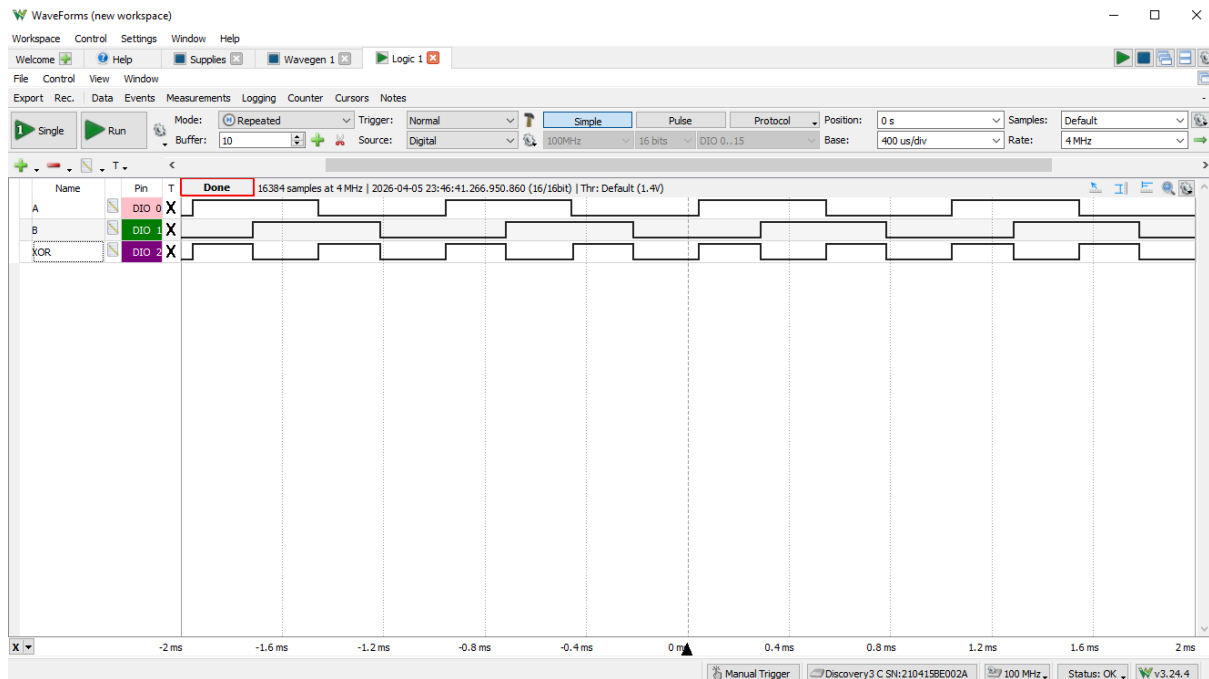
Functional Testing

To verify the functional correctness of the CMOS XOR gate, the circuit was tested to ensure its output accurately reflected the standard XOR truth table. The waveform generator was configured to supply two square waves from 0-5V, set with a 2.5V amplitude and a 2.5V offset. By introducing a 90-degree phase shift to one of the input signals, the setup automatically cycled through all four binary combinations (00, 01, 11, and 10). The logic states were monitored using the Digital I/O pins on the AD3 using the Logic Workspace on Waveforms.

Table 1: Truth Table of the XOR Gate

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

Figure 3: Logic Workspace test of the CMOS XOR gate for the Functional Test.



From viewing the Logic Workspace in Waveforms with the voltages applied to the CMOS XOR gate, the output correctly functions as a XOR gate.

Static Level Testing

Static testing was conducted by holding one input at a constant +5V (logic high) while driving the second input with a square wave from 0-5V. The roles of the inputs were then swapped to observe any discrepancies caused by the different physical paths through the circuit. When Input A was held high, the measured output levels were $V_H = 5.0455 \text{ V}$ and $V_L = 16.055 \text{ mV}$. After swapping the inputs to hold B high, the output levels shifted to $V_H = 5.0437 \text{ V}$ and $V_L = 14.280 \text{ mV}$.

The slight variations observed between the two configurations can be caused by differences in the internal resistance of the transistor paths, breadboard resistance, or slight impedance from the jumper wires.

Overall, the percent error is negligible as V_H and V_L for either static test was well within the expected range. For example, using the measured value for Input A yielded a percent error of 0.91% for V_H , indicating that there is very little error.

$$\% \text{ Error of } V_{H,A} = \frac{|5.0455 - 5|}{5} \times 100\% = 0.91\%$$

Figure 4: Scope with A set to a square wave from 0-5V and B set to a 5 VDC source. The orange waveform represents the XOR output, while the blue waveform represents A.

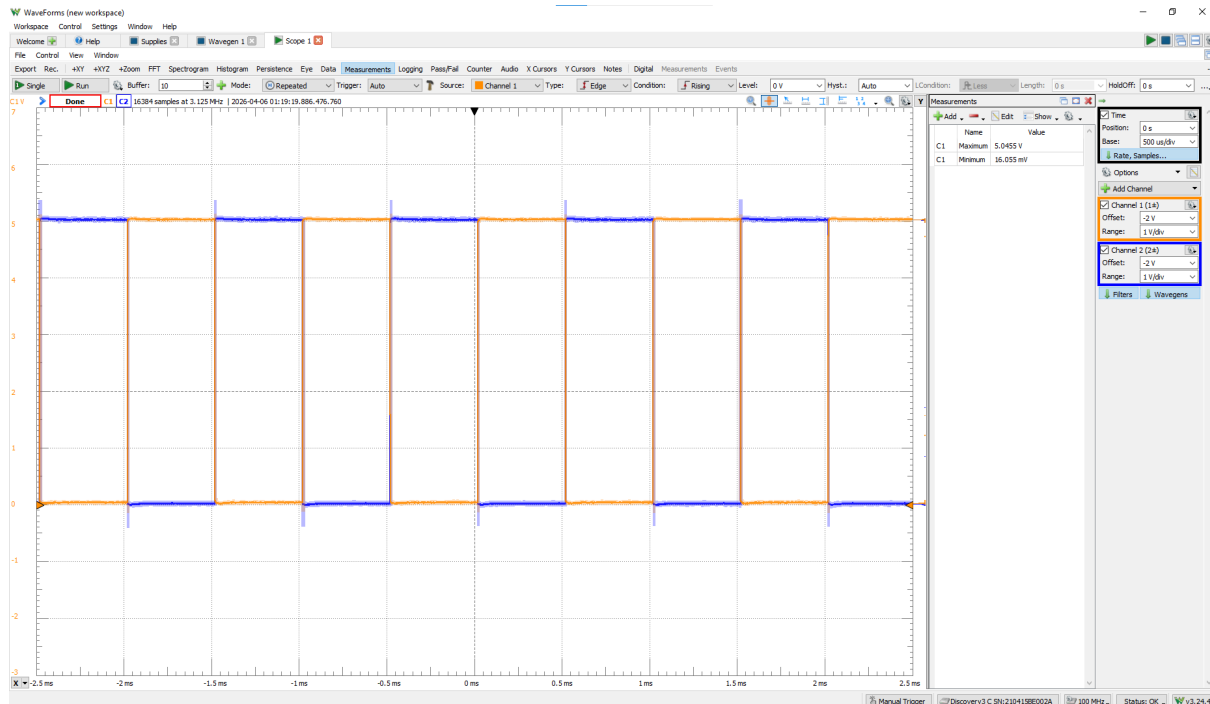
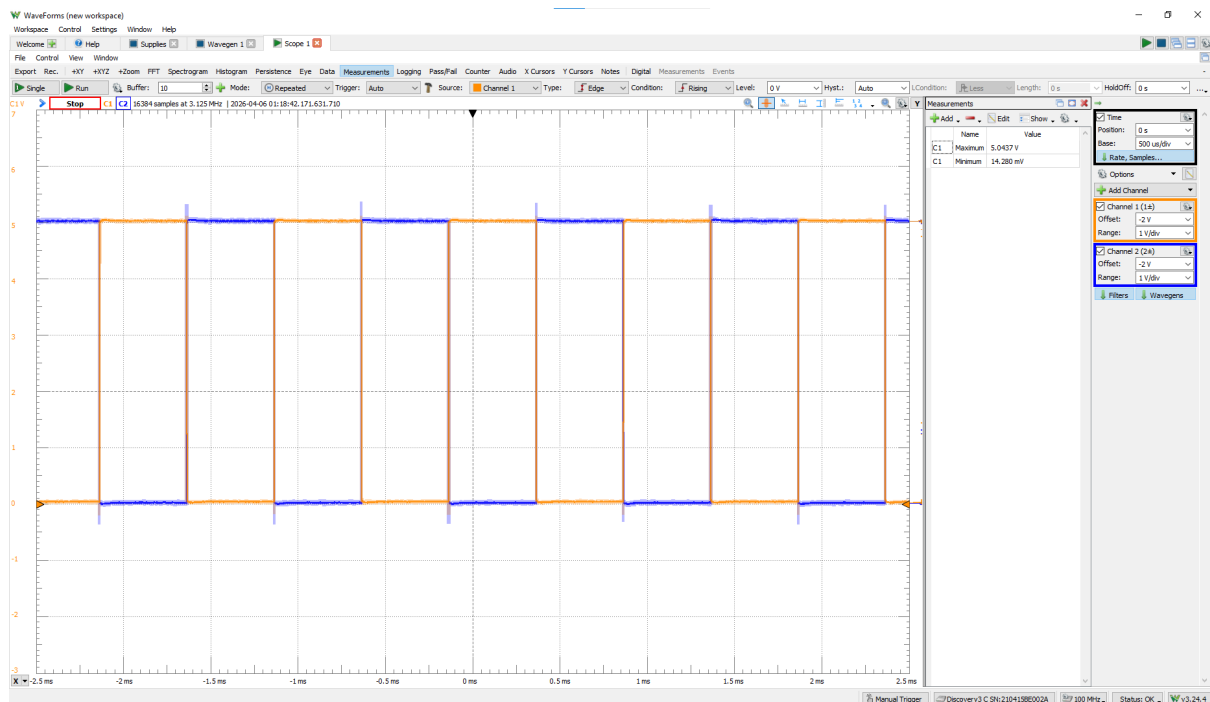


Figure 5: Scope with B set to a square wave from 0-5V and A set to a 5 VDC source. The orange waveform represents the XOR output, while the blue waveform represents B.

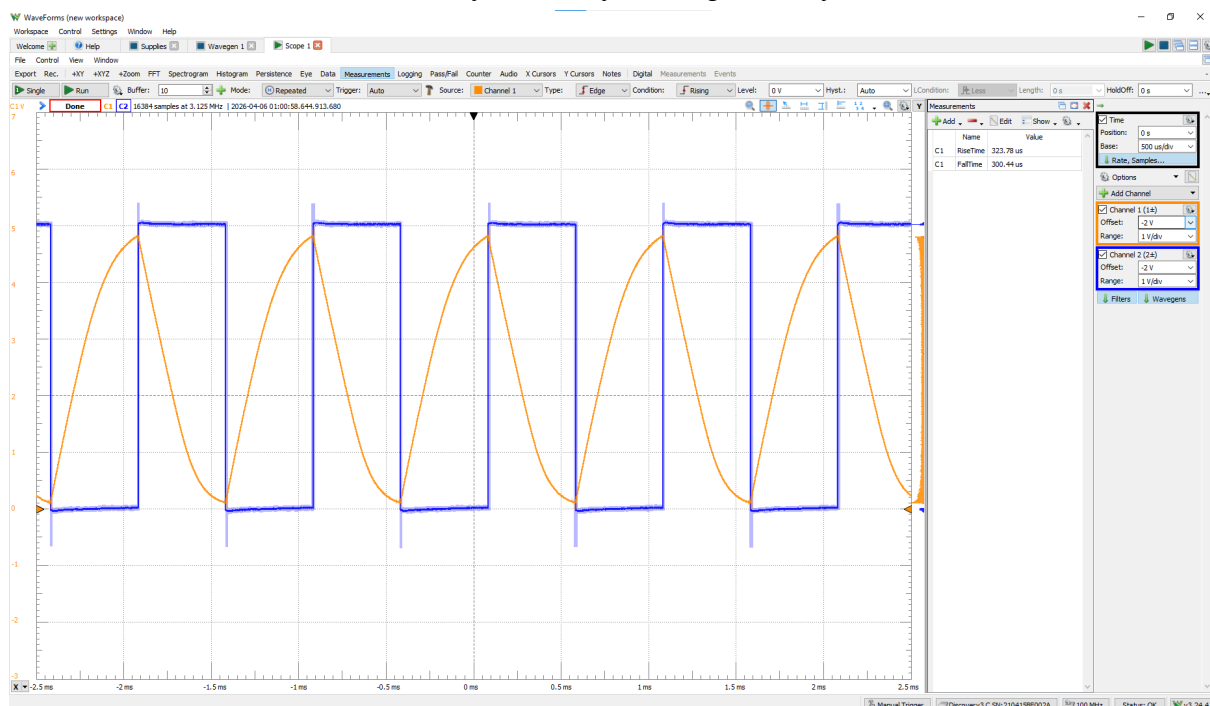


Timing

Rise and Fall Times

To evaluate the switching performance of the circuit, a timing analysis was conducted to determine the transition delays between logic states. This was achieved by connecting a 100 nF capacitor to the output node to simulate a load and measuring the resulting voltage across it. During the test, input B was held at a constant 5V DC, while input A was driven by a 1 kHz square wave with a 2.5V amplitude and a 2.5V offset. From the captured waveform, the rise time was measured as $t_{\text{rise}} = 323.78 \mu\text{s}$ and the fall time as $t_{\text{fall}} = 300.44 \mu\text{s}$.

Figure 6: Scope measuring the circuit with a 100 nF load for timing, specifically measuring the rise and fall time of the output waveform.



Propagation Delay Timings

To determine the high-to-low propagation delay (τ_{PHL}), the time difference was measured from the 50% voltage mark of the input transition to the 50% voltage mark of the output's falling edge, resulting in $\tau_{\text{PHL}} = 162.226374 \mu\text{s}$.

To determine the low-to-high propagation delay (τ_{PLH}), the time difference was measured from the 50% voltage mark of the input transition to the 50% voltage mark of the output's rising edge, resulting in $\tau_{\text{PLH}} = 162.74049 \mu\text{s}$.

Figure 7: Scope measuring the circuit with a 100 nF load for timing, specifically measuring τ_{PHL} using the X-Cursors. $\tau_{PHL} = 162.226374 \mu\text{s}$.

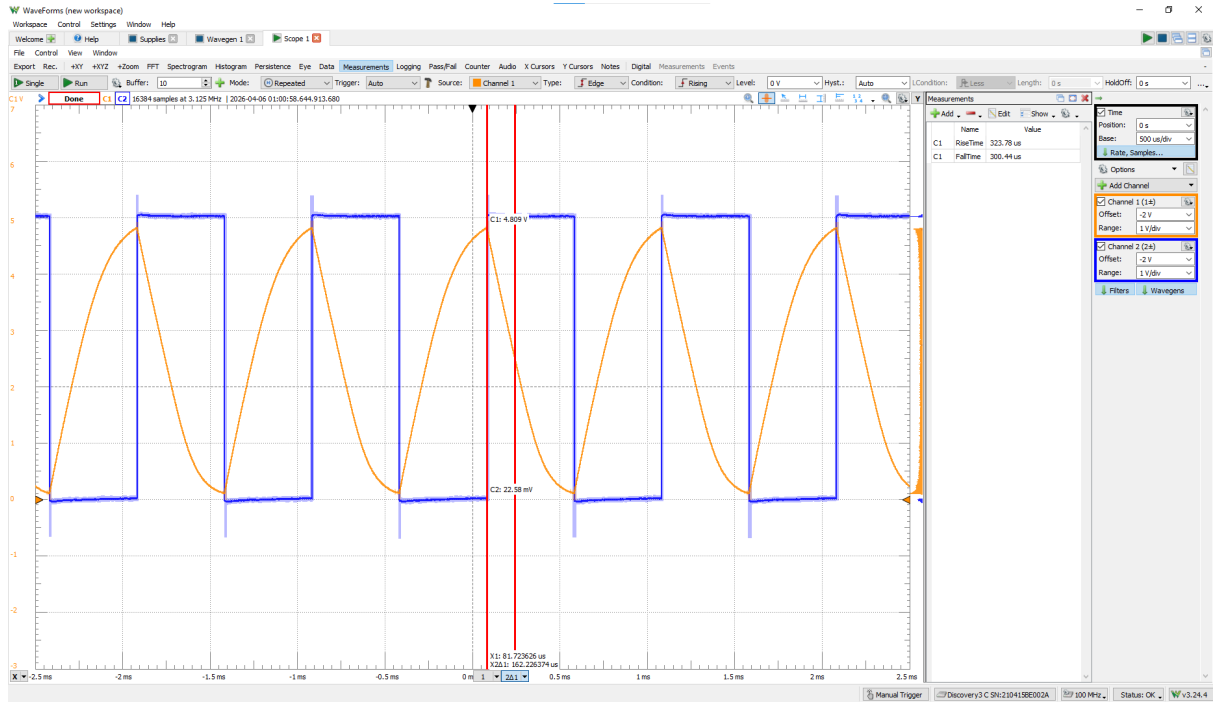
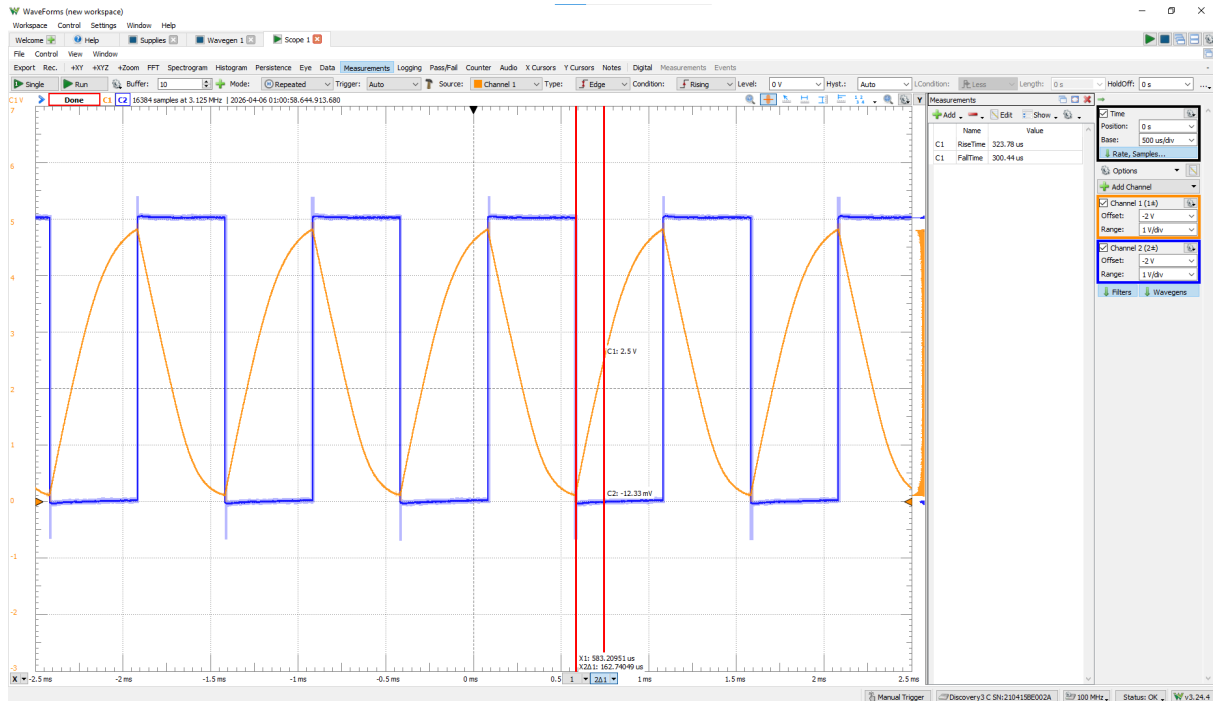


Figure 8: Scope measuring the circuit with a 100 nF load for timing, specifically measuring τ_{PLH} using the X-Cursors. $\tau_{PLH} = 162.74049 \mu\text{s}$.



Using these values, the overall average propagation delay (τ_p) was calculated as follows:

$$\tau_p = \frac{\tau_{PHL} + \tau_{PLH}}{2} = \frac{162.226374 + 162.74049 \mu\text{s}}{2} = 162.483432 \mu\text{s}$$